KAFANDO THOMAS

MECHANICS AND ROBOTICS ENGINEER

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Mohammedia, Morocco
My LinkedIn Profile
My Portfolio

PROFILE

I'm a Robot Mechanical Modeling and Simulation Engineer with a Master's degree in Mechanics, Robotics, and Innovative Materials. My thesis internship gave me hands-on experience in mechanical modeling and numerical simulation, and I also have solid practical experience with the ROS 2 navigation stack and MoveIt 2 for robot motion planning. "Serving, my happiness. Innovating, my determination".

SOFT SKILLS

- Teamwork, project lead,Time Management,Critical Thinking
- Effective Communication,

TECH SKILLS

- systems control , control theory
- Python , C++,Matlab/Simulink
- Data Structures and Algorithms
- ROS2,gazebo,nav2,MoveIt2,ompl
- Tensorflow (Keras) and scikit-learn
- mechanical systems dynamics modeling
- Deep Learning literature and training
- MPC, Sliding control

LANGUAGES

- French (Native),Moore (Native)
- English (Upper Intermediate)
- Arabic (beginner)

CERTIFICATIONS

- FREE TensorFlow-Keras Bootcamp
- Introducing Generative AI with AWS
- ROS2 FOR BEGINNERS
- AWS Educate Machine Learning Foundations
- ROS2 FOR BEGINNERS Level 2
- ROS2 FOR BEGINNERS Level 3
- ROS2 - Hardware and ros2_control
- MATLAB Onramp

HOBBIES

- Football, History reading
- Wandering on mountains

REFEREES

- Professor ABDELLAH AILANE, my teacher of Advanced Systems Control
  abdellah.ailane@enset-media.ac.ma
  +212 666-136103
- Professor ABDERRAHIM BAHANI, my teacher of Industrial Robotics
  a.bahani@enset-media.ac.ma
  +212 679-745676

EDUCATION

Master of Mechanics, Robotics and innovative Materials
Hassan II University of Casablanca - ENSET of Mohammedia 2024 - 2026 *grade : Good*

BSc. of Mechanical Engineering
Sultan Moulay Slimane University - FST of Beni Mellal 2023 - 2024 *grade : Good*

University Diploma of Science and technology : Mathematics, computer science, Physics and Chemistry
Sultan Moulay Slimane University - FST of Beni Mellal 2021 - 2023 *grade : Good*

EXPERIENCES

Robot modeling and simulation engineer MAR-AUG2024
*Laboratory of Research : **M**odélisation et **S**imulation des **S**ystèmes **I**ndustriels **I**ntelligents-M2S2I*

I designed and modeled the robot in CAD and URDF, implemented autonomous navigation and manipulation using Nav2 and MoveIt 2, and built the decision-making architecture with BehaviorTree.CPP. Currently, I am transitioning to real hardware deployment.
Impact: streamlined rubber tree latex collection and significant productivity enhancement in agricultural operations.
Technologies Used: Solidworks, Blender, ROS 2, Nav2, MoveIt 2, BehaviorTree.CPP, Latex

Mechanical design intern APR-MAI 2024
Company/Service : Service de Logistique et du Matériel | Beni Mellal

During that internship we have been working on the study of feasibility and the design of a snow plow. I have learnt from it, mechanical modeling and simulation. I personally grasped the challenge of moving from theoretical modeling to numerical modeling.
Technologies Used: SOLIDWORKS, ANSYS, Microsoft Office

PROJECTS

Simulation of temperature in an heated room OCT-DEC 2024
ENSET Mohammedia | 2024-2025
During this project, we firstly found the theoretical modeling with differential equations based on conduction and convection. Next, We moved on to munerical modeling using Euler descretization techniques to simulate the heat exchange .
Technologies Used: Matlab/simulink and Microsoft Office

Particle filter implementation in 2D
ENSET Mohammedia | 2024-2025
The motivation for this project was a personal curiosity to better understand both the theory and the practice. In this project, I have been able to simulate the trajectory estimation of a simple mass-spring system using an object-oriented approach in Python.
Technologies Used: Python

Industrial Production Pipeline optimatization with machine learning
ENSET Mohammedia | 2025-2026
The missions accomplished in that project are:

- Development of a predictive model pipeline using Scikit-learn to optimize industrial throughput by analyzing real-time sensor data integrated via the MQTT protocol.
- Engineering of a seamless data flow from hardware sensors to mobile phone dynamic visualizations, enabling proactive bottleneck identification and process efficiency gains.

Technologies Used: scikit-learn library, vs code, MQTT , ...